GROUP WORK PROJECT # 2
GROUP NUMBER: 13091

MScFE 660: RISK MANAGEMENT

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
SK Wasim Ahmed	India	ahmedw47@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
Team member 1	**SK Wasim Ahmed**
Team member 2	
Team member 3	

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

INTRODUCTION

The project focus on methodological design and model development required to forecast West Texas Intermediate (WTI) crude oil prices. As identified in previous group project financial time series data exhibits non stationarity complex dependencies between global macroeconomic factors and market prices.

METHODOLOGY

The project involves use of hidden markov models to perform regime price detection. The goal is to identify the hidden states namely Bull Bear and Stagnant regimes based on sequence of oil price returns. This is achieved by implementing Forward, Backward, Viterbi, and Baum-Welch algorithms. To implement the frameworks python programming is used with specific libraries hmms and pgmpy. The report details the mathematical pseudocode for algorithms validating them the toy examples and outline the design process for models.

STEP 1

Toy Example Definition

Hidden States (Q): q_1: Bull Market q_2: Bear Market Number of hidden states: N = 2

Observations (V): v_1: Price Up v_2,: Price Down

Observation Sequence:

O = (o_1, o_2), Length of the sequence: T = 2

Model Parameters ($\lambda = (A, B, \pi)$):

Initial State Probabilities (π): Probability of Bull Market (π_{bull}) = 0.6 Probability of Bear Market (π_{bear}) = 0.4

State Transition Probabilities (A):

Probability of transitioning from Bull Market to Bull Market ($a_{bull \to bull}$) = 0.7

Similarly ($a_{bull \to bear}$)= 0.3 ($a_{bear \to bear}$) = 0.8 ($a_{bear \to bull}$) = 0.2

Emission Probabilities (B):

In Bull Market:

$b_{Bull}(Up)$ = 0.9 $b_{Bull}(Down)$ = 0.1

In Bear Market:

$b_{Bear}(Up) = 0.3$

$b_{Bear}(Up) = 0.7$

Student A

Pseudocode for the forward algorithm and the backward algorithm

$\alpha_t(j)$ is the probability of being in state j after seeing the first t observations:

```
FUNCTION Forward(O, lambda = (A, B, pi)):
  N = number of states
  T = length of O
  create a matrix alpha[N, T]
  # Initialization
  FOR each state s from 1 to N:
    alpha[s, 1] = pi[s] * b[s](o[4])
  # Recursion
  FOR each time step t from 2 to T:
    FOR each state s from 1 to N:
      sum_prev = 0
      FOR each state s_prev from 1 to N:
        # Summing probabilities from all previous states
        sum_prev = sum_prev + (alpha[s_prev, t − 1] * a[s_prev, s])
      alpha[s, t] = sum_prev * b[s](o[t])
```

$\#\ Termination$

$P(O|lambda)\ =\ sum(alpha[s,T]\ for\ s\ from\ 1\ to\ N)$

$RETURN\ alpha, P(O|lambda)$

Backward Algorithm: Computes the probability of future observations from $t\ +\ 1$ to T given state i at time t. We define the backward variable β.

$FUNCTION\ Backward(O, lambda\ =\ (A, B, pi))$:

$N\ =\ number\ of\ states$

$T\ =\ length\ of\ O$

$create\ a\ matrix\ beta[N,T]$

$\#\ Initialization\ (End\ of\ sequence)$

$FOR\ each\ state\ s\ from\ 1\ to\ N$:

$beta[s,T]\ =\ 1$

$\#\ Recursion\ (Going\ backwards\ from\ T-1\ to\ 1)$

$FOR\ each\ time\ step\ t\ from\ T-1\ down\ to\ 1$:

$FOR\ each\ state\ i\ from\ 1\ to\ N$:

$sum_future\ =\ 0$

$FOR\ each\ state\ j\ from\ 1\ to\ N$:

$\#\ Transition\ i->\ j\ *\ Emission\ in\ j\ *\ Future\ Probability\ from\ j$

$sum_future\ =\ sum_future\ +\ (a[i,j]\ *\ b[j](o[t+1])\ *\ beta[j,t+1])$

$beta[i,t]\ =\ sum_future$

$RETURN\ beta$

Toy example calculation:

Forward at $t = 1$

$$\alpha_1(1) = \pi_1 * b_1(Up) = 0.6 * 0.9 = 0.54$$

$$\alpha_1(2) = \pi_2 * b_2(Up) = 0.4 * 0.3 = 0.12$$

Backward at $t = 1$

We sum over next states $j = 1$ (Bull) $j = 2$ (Bear) at $t = 2$

$$\beta_1(1) = a_{11} b_1(Down) \beta_2(1) + a_{12} b_2(Down) \beta_2(2)$$

$$\beta_1(1) = (0.7 * 0.1 * 1) + (0.3 * 0.7 * 1) = 0.07 + 0.21 = 0.28$$

STUDENT B

Viterbi Algorithm uses dynamic programming in finding most probable sequence of states.

$FUNCTION\ Viterbi(O, lambda = (A, B, pi))$:

$N = number\ of\ states$

$T = length\ of\ O$

$create\ a\ path\ probability\ matrix\ viterbi[N, T]$

$create\ a\ backpointer\ matrix\ backpointer[N, T]$

$\#\ Initialization$

$FOR\ each\ state\ s\ from\ 1\ to\ N$:

$viterbi[s, 1] = pi[s] * b[s](o[4])$

$backpointer[s, 1] = 0$

$\#\ Recursion$

$FOR\ each\ time\ step\ t\ from\ 2\ to\ T$:

```
  FOR each state s from 1 to N:
    # Find max probability from previous states s_prime
    max_prob = -1
    best_prev_state = 0
    FOR s_prime from 1 to N:
      prob = viterbi[s_prime, t - 1] * a[s_prime, s] * b[s](o[t])
      IF prob > max_prob:
        max_prob = prob
        best_prev_state = s_prime
    viterbi[s, t] = max_prob
    backpointer[s, t] = best_prev_state
# Termination
bestpathprob = max(viterbi[s, T] for s in 1..N)
bestpathpointer = argmax(viterbi[s, T] for s in 1..N)
# Backtrace
bestpath = list()
current_state = bestpathpointer
add current_state to bestpath
FOR t from T down to 2:
  current_state = backpointer[current_state, t]
```

add current_state to bestpath (at beginning)

RETURN bestpath, bestpathprob

Toy Example Calculation:

Initialization ($t = 1$):

viterbi=0.54 (Bull)

viterbi=0.12 (Bear)

Recursion ($t = 2$) finding max path to State 2 (Bear) given $\sigma_2 = Down$

From Bull (q_1): $viterbi * a_{12} * b_2(Down) = 0.54 * 0.3 * 0.7 = 0.1134$

From Bear (q_2): $viterbi * a_{22} * b_2(Down) = 0.12 * 0.8 * 0.7 = 0.0672$

Result: $viterbi = 0.1134$. The backpointer points Bull (q_1)

If viterbi is the max final probability, the path is Bull → Bear

STEP 2

Regime identification

STUDENT A (Bull Regimes) (H_1): The bull regime is a latent state where consistent upward trend of asset observed. In this state following properties are observed:

Positive Mean ($\mu > 0$): Positive expected daily returns.

Moderate Variance (σ^2): bull markets in oil tend to have lower volatility.

Based on the previous group project (GWP1) data WTI Crude Oil Price dataset in the periods of (1990–2023) the bull regime occurred in periods of 2010 and mid 2014

Post 2008 Financial Crisis WTI crude oil recovered from ~$40/bbl to stabilize above $100/bbl for several years. Viterbi algorithm in this period primarily decode hidden state as H_1 i.e bull. Although returns were consistently positive but in small proportion absence of the volatility component observed.

STUDENT B (Bear Regimes) (H_2): bear regimes correspond to market contraction. This state is also mathematically distinct as it is able to capture the market behavior. The following characteristics are:

Positive Mean ($\mu < 0$): Negative expected daily returns.

High Variance (σ^2): bull markets in oil tend to have lower volatility.When oil markets turn bearish prices usually drop fast. supply shocks or demand collapse is observed. HMM is able to detects this state through the spike in variance.

Illustrations:

In Global financial crisis Oil prices collapsed from $147 to $32 in under six months.

The 2020 COVID-19 Crash: This implies High Volatility Bear State where standard deviation of emission probability distribution maximized.

STEP 3

Definition of Hidden Markov Model

Source Reference: Jurafsky, Daniel, and James H. Martin. Speech and Language Processing. 3rd ed., Pearson, 2024

A Hidden Markov Model (HMM) is a probabilistic sequence model that computes a probability distribution over possible sequences of labels and chooses the best label sequence. Unlike a standard Markov Chain, where each state corresponds to an observable event, an HMM models a system where the states are "hidden" (latent) and only the emissions (observations) generated by those states are visible to the observer.

HMM is defined as a 5-tuple $\lambda = (Q, A, O, B, \pi)$

Set of Hidden States (Q): A set of N states Q = $\{q_1, q_2, \ldots, q_N\}$ these actually represent the latent market regimes.

Transition Probability Matrix (A): A matrix $A = [a_{ij}]$ a_{ij} is probability of moving from state i to j

$$a_{ij} = P(q_t = j | q_{t-1} = i), \quad 1 \leq i, j \leq N$$

Set of Observations (O): A sequence of T observed symbols O = $\{o_1, o_2, \ldots, o_T\}$

drawn from a vocabulary V

Emission Probability Matrix (B): A set of observation likelihoods which is also called emission probabilities, where $B = [b_j(o_t)]$ probability of an observation o_t being generated from a state j

$$b_j(o_t) = P(o_t|q_t = j)$$

Initial Probability Distribution (π): A vector $\pi = [\pi_1, \pi_2, ..., \pi_N]$ probability that the system starts in state i at time t=1

$$\pi_i = P(q_1 = i)$$

The following assumptions are stated below:

1. The probability of a future state depends only on the current state, not on the sequence of events that preceded it.

$$P(q_t|q_{t-1}, q_{t-2}, ..., q_1) \approx P(q_t|q_{t-1})$$

2. The probability of an observation o_t depends on the state that produced it (q_t) and is independent of all other observations and states.

$$P(o_t|q_T, ..., q_1, o_T, ..., o_1) \approx P(o_t|q_t)$$

STEP 4

Design Methodology

Macroeconomic Indicator Identification (Student A): Formulation of dataset on WTI spot price done model risk dynamics.
Three primary categories of indicators identified for dataset:

1. Global Crude Oil Production and US Inventory levels indicate reality of markets. downward pressure on prices exerted by the surplus in physical inventory. This is indicator of bear regime.
2. US Dollar Index (DTWEXBGS)included because oil is priced in USD. Nominal price of oil when dollar appreciates and vice versa.
3. St. Louis Fed Financial Stress Index (STLFSI) selected as financial stress indicator as acts as a proxy for systemic risk. High values in this index correlate with liquidity crisis as result this force traders to liquidate positions in commodities causing crash states.

Data Cleaning and Sterilization Protocol:

To prepare this data for the HMM we applied a rigorous sterilization protocol i.e cleaning data:

1. We had to change the daily spot prices to a monthly (Month End) frequency. We did this sothat the financial figures would match up with how often less frequent monthly economic stats like Industrial Production are reported. Basically we got the fast moving financial data to sync with reporting cycle of the macroeconomic statistics.
2. Using Forward-Fill method missing values handled. As volatility is the primary signal HMM uses to detect regimes preserving the roughness of the original time series.
3. The core assumption of the standard hidden markov models with gaussian emissions is that statistical properties of the data are constant within a given state. So the prices are converted to log returns which is r_t

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Regime Detection Process (Student B):

A single underlying process under traditional time series data and its postulated that the oil market switches between two distinct stochastic processes:

1. Regime 0 (Bull/Stable): steady demand growth, positive expected returns, and low volatility.
2. Regime 1 (Bear/Crisis): market panic, negative expected returns, and high volatility.

Gaussian Hidden Markov Model implemented which assumes assumes that the observable data emitted from a Gaussian distribution whose parameters (Mean μ and Variance σ^2) depend entirely on the current hidden state q_t.

Code implementation: hmm python library used and the specific class employed was GaussianHMM.
Model implementation: The model restricted to two states to maintain interpretability (Bull vs. Bear).
Covariance type: The parameter set to "full" to instruct the model to learn a separate and independent covariance matrix for each hidden state.
Training phase: utilizes the Baum-Welch Algorithm.

1. The algorithm calculates the probability of being in State i at time t ($\gamma_t(i)$) given the current estimates of the parameters. It uses the Forward-Backward algorithm for smoothing over entire timeline.
2. M-Step (Maximization): updates the model parameters (π, A, μ, Σ) to maximize expected likelihood of the observed data.
3. Convergence: specified threshold below 0.01 used for stabilization.

Decoding phase: Viterbi used in computation the single most probable path of hidden states $q_1, q_2, \ldots, q_T$ maximizing joint probability $P(Q, O|\lambda)$

Analysis of Learned Parameters:

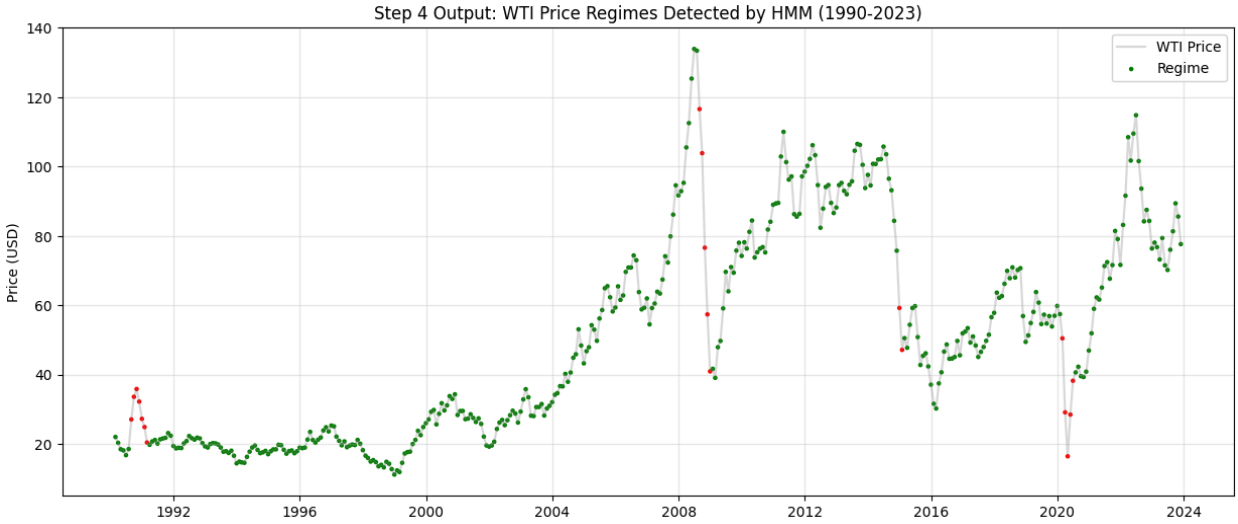


Fig 1: WTI Price Regimes Detected by HMM (1990-2023)

State 0 Identification: low value of variance of 0.00534 indicates a tight distribution of returns implying a state of Bull regime.

State 1 Identification:model learned variance of 0.06545. The variance in State 1 is approximately 12.2 times higher than in State 0. The high variance state effectively captures the fat tails of the return distribution and market crashes.

STEP 6

Student A: Time Series Transformation and Stationarity

In our analysis of WTI spot price from 1990 to 2023 it is revealed crude oil prices are inherently nonstationary. The data follows a random walk and structural break. Using raw prices directly into the HMM not feasible as it would fail to detect the underlying volatility regimes as it would group data points by price level.

raw spot prices (P_t) into Log-Returns (r_t) converted as it would oscillate around a stable mean of approximately zero

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Student B

The task is to infer the most likely sequence of hidden market regimes ($Q = q_1, \ldots, q_T$) once the model parameters $\lambda = (A, B, \pi)$ are learned via the Baum Welch algorithm. This helps in explaining observed history of oil returns.

Viterbi Algorithm is used in reconstructing optimal path the algorithm operates in two passes:

1. Forward Pass: Algorithm moves forward through time intervals in periods from 1990 to 2023. At each time step t calculating probability of arriving at a specific state (whether bull or bear state) given the most likely path leading up to it. It stores a backpointer to the previous state that maximized probability.
2. Backward Pass: The algorithm then identifies the final state with highest probability once it reaches the end of the period. It then follows backpointers in reverse order in reconstructing single global path maximizing joint probability.

Step 7

we have identified two distinct regimes on our model from the step 4:

State 0: The Stable Regime (Bull Market): model identified this state with a variance of $\sigma^2 \approx 0.005$ implying relatively balanced supply and demand 2010–2014 recovery to be more specific.

State 1: The Crisis Regime (Bear Market): model identified this state with a variance of $\sigma^2 \approx 0.065$ indicating periods of systemic shock and risk off sentiment.

Analysis of Learned Transition Matrix

The Baum Welch algorithm successfully estimated the State Transition Matrix (A) that helps in governing the dynamics of regime switching.

$$\mathbf{A} = \begin{bmatrix} P(\text{State } 0 \to \text{State } 0) & P(\text{State } 0 \to \text{State } 1) \\ P(\text{State } 1 \to \text{State } 0) & P(\text{State } 1 \to \text{State } 1) \end{bmatrix} = \begin{bmatrix} 0.98412717 & 0.01587283 \\ 0.22914644 & 0.77085356 \end{bmatrix}$$

High Persistence of Stability (State 0): For State 0 model learned a self-transition probability of 98.4% indicating Bull/Stable regime as default state of the oil market.

State 1: For State 1 self transition probability 77.1% confirming that high volatile regimes are sticky. Once crisis begins it persists or an extended period. Thus it validates the use of HMMs over simple static models as it captures time dependent nature of market risk.

Step 9

Methodology

Hill Climb Search algorithm is implemented using the pgmpy library. This is useful in determining the causal dependencies between our macroeconomic variables and the WTI Crude Oil price. Hill Climb Search is a score based structure learning algorithm useful in finding the Directed Acyclic Graph (DAG) that best fits the data. Bayesian Information Criterion (BIC) used as scoring metric rewards model accuracy while penalizing number of edges. This helps in prevention of overfitting training data.

Algorithm process:

The algorithm begins with no edges (disconnected graph). In each iteration the algorithm considers all possible single edge operations such as adding an edge ($X \to Y$), deleting an edge($X \nrightarrow Y$), or reversing an edge($Y \to X$). Any operation that creates a cycle (for e.g $A \to B \to A$) discarded to ensure result as DAG. The score change ($\Delta Score$) is then calculated by the algorithm for every valid operation. The loop is repeated until no single operation improves the score implying the search has reached a local maximum.

Bayesian Information Criterion (BIC):

BIC score is composed of two competing terms:

$$BIC(\mathcal{G} : \mathcal{D}) = \underbrace{\mathcal{L}(\mathcal{G} : \mathcal{D})}_{\text{Likelihood}} - \underbrace{\frac{\log M}{2} Dim(\mathcal{G})}_{\text{Penalty}}$$

Penalty Term: This penalizes score based on the sample size (M) and the number of parameters ($Dim(\mathcal{G})$)

Likelihood Term ($\mathcal{L}$): This measures how well the graph explains the observed data

Visual Analysis

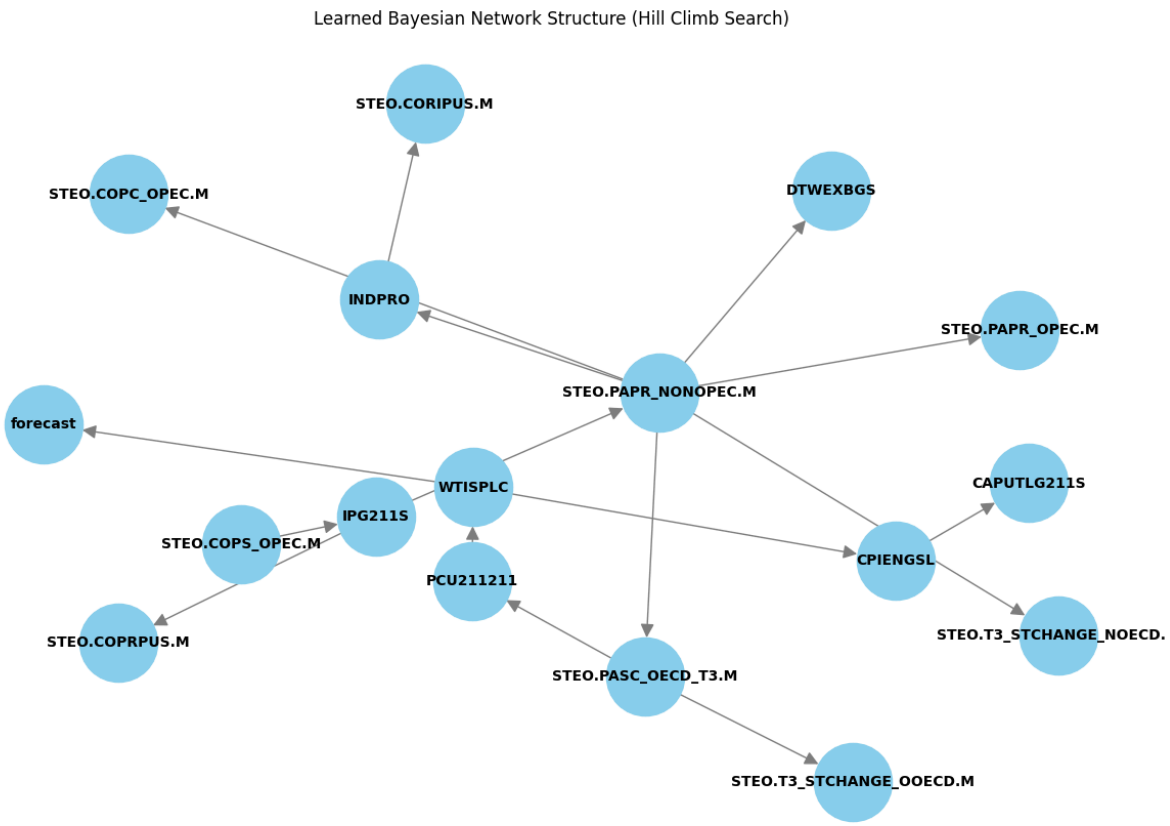


Fig 2: Learned Bayesian Network Structure (Hill Climb Search)

The model has "learned" that Non-OPEC supply is the dominant independent variable in this dataset influencing both OPEC policy and industrial output. The link from Producer Prices (PPI) to WTI Spot Price indicate fundamental cost basis for pricing while link from WTI Price to Forecast confirms that market regimes are defined by observable price volatility.

Results & Economic Interpretation

Learned Edge (Cause → Effect)	Economic Interpretation
PCU211211 → WTISPLC	PPI for crude petroleum identified as a parent of the spot price. This indicates a costpush relationship where the fundamental costs of extraction and production are direct drivers of the market price.
STEO.PAPR_NONOPEC → STEO.PAPR_OPEC	This captures modern market dynamics (e.g., the U.S. Shale boom) where supply surges from non cartel members force OPEC to adjust their quotas defend market share or support prices.
WTISPLC → forecast	The model identifies the spot price as parent of the forecast (Regime). This implies that price level and its volatility are observable signals that help in defining the market state.Extreme price movements reveal whether regime is in bull or bear

Latent Meaning Behind Hidden States:

Based on step 6 two latent states identified:

State 0: The "Bull"/Stable Regime: Market uncertainty is low in this regime implying lower volatility states with steady mean returns.

State 1: The "Bear"/Volatile Regime: This state correlate with geopolitical uncertainty such as 2008 Financial Crisis and 2020 COVID-19 demand shock.

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